

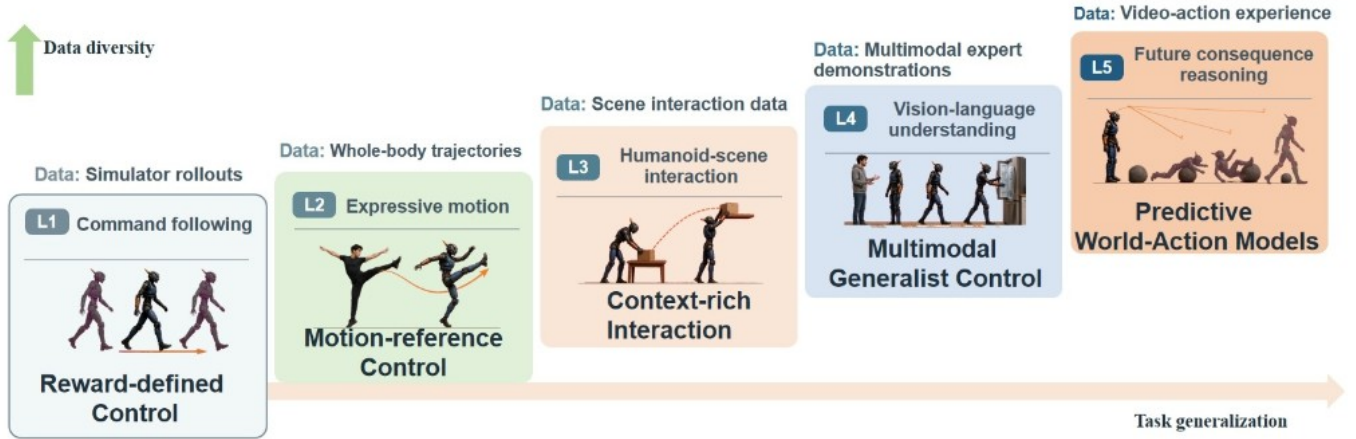


Fig. 1. Evolution of learning paradigms for humanoid whole-body control along the axis of data diversity and task generalization. Existing approaches progressively evolve from reward-defined learning with simulator rollouts, through motion-conditioned and context-grounded interaction learning, toward multimodal generalist control and predictive world-action modeling. The transition reflects an increasing richness of task specification, from manually designed objectives to multimodal intent and future consequence reasoning.

I. LEARNING-BASED WHOLE-BODY CONTROL

Learning-based humanoid control has evolved from reward-defined task learning toward open-world generalist control as data priors become increasingly diverse and expressive. As summarized in Fig. 1, this progression moves from self-generated simulation rollouts to expressive motion imitation, real world interaction, multimodal reasoning, and predictive world understanding, enabling a gradual shift from command following toward general motion, physical interaction, semantic grounding, and future-aware decision making.

A. Robust Command Following

At the low-data-diversity end of κ_d , humanoid control can be learned through pure reinforcement learning (RL) without external behavioral demonstrations. A policy $\pi_\theta(a_t | o_t, c_t)$ learns directly from environment interaction, with its training data generated by its own rollouts. Task intent is specified through hand-designed rewards and low-dimensional commands c_t , such as desired base velocity, heading, foothold, or task-space targets. This paradigm has progressed from basic locomotion to challenging terrain traversal, parkour, and sparse-foothold locomotion [1–5]. Despite the simple task intents, the policy itself may exploit rich input information such as observation histories or learned state representations [15–17].

The practical success of this regime is closely tied to simulation scale. High-dimensional actuation, underactuated balance, discontinuous contacts, and costly hardware failures make direct real-world exploration impractical, so modern humanoid RL is typically trained in massively parallel GPU simulation [1, 2]. Online RL algorithms like PPO remain the mature default because of its stable optimization and compatibility with large-scale parallel rollouts [1, 2, 11]. Recent off-policy methods such as FastTD3 and FlashSAC, however, show that replay-based learning can also scale effectively to high-dimensional humanoid control when critic stability and update scale are carefully managed [9, 10].

Beyond the simulation environments and RL algorithm regimes, practical humanoid RL relies on several other training techniques to ensure stable optimization and deployment robustness. Curriculum learning progressively expands command, terrain, or contact difficulty, while privileged critics or simulator-only state information provide oracle supervision during training without changing the deployed policy interface [3, 4]. Sim-to-real transfer is typically supported by domain randomization, system identification, and calibration, while residual mismatches can be further corrected through limited real-world adaptation [1, 2, 12]. At deployment, joint-position or joint-offset targets tracked by low-level PD controllers remain the dominant action interface, providing a hardware-compatible control space while retaining fast local feedback [1, 2, 13].

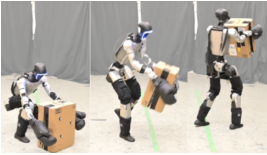
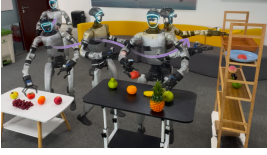
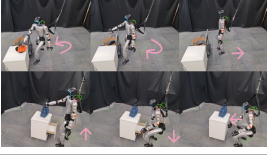
Reward-defined learning works well when task success can be captured by compact, measurable objectives. Commands such as target velocity or foothold location fit this setting naturally, but richer behaviors such as dancing, martial arts, and other highly coordinated motions are much harder to specify with hand-crafted scalar rewards. The bottleneck therefore shifts from simply learning the task to specifying what the desired behavior should look like. Human motion demonstrations provide a richer data prior for this purpose, offering trajectory-, distribution-, or representation-level guidance for physically grounded learning [29, 39]. This shift from policy-generated rollouts to external motion data marks the first major expansion along κ_d and motivates the motion-conditioned regime that follows.

B. Expressive Whole Body Motion

a) *Motion Data Collection and Robot Alignment:* Compared with pure reward-defined control, demonstration motion data offer a richer task intent by directly describing coordinated whole-body behavior. Large-scale offline motion corpora are commonly obtained from motion-capture datasets such as AMASS, CMU, and SFU, or reconstructed from human videos

TABLE I

LITERATURE LANDSCAPE OF LEARNING-BASED HUMANOID WHOLE-BODY CONTROL. STUDIES CITED THROUGHOUT THIS CHAPTER ARE ORGANIZED BY THEIR DOMINANT ROLE IN THE PROGRESSION FROM ROBUST COMMAND FOLLOWING TO PREDICTIVE WORLD-ACTION MODELS; SEVERAL WORKS NATURALLY SPAN MULTIPLE PARADIGMS.

Paradigm	Illustration	Literature covered in this chapter
L1 Robust command following		Real-World Humanoid Locomotion [1]; 15-min Sim2Real Humanoid Locomotion [2]; Humanoid Parkour [3]; BeamDojo [4]; Learning Humanoid Locomotion over Challenging Terrain [5]; Learning Perceptive Humanoid Locomotion [6]; Humanoid-Gym [7]; Booster Gym [8]; FastTD3 [9]; FlashSAC [10]; PPO [11]; ASAP [12]; HOVER [13]; FALCON [14]; Memory-Based Bipedal Control [15]; Rapid Motor Adaptation for Bipeds [16]; Perceptive Internal Model [17].
L2 Expressive whole body motion		AMASS [18]; CMU MoCap [19]; SFU MoCap [20]; HumanPlus [21]; H2O [22]; OmniH2O [23]; TWIST [24]; TWIST2 [25]; RoboMirror [26]; Retargeting Matters [27]; Human2Humanoid [28]; DeepMimic [29]; HiLo [30]; UniTracker [31]; GMT [32]; Adaptive Motion Tracking [33]; KungfuBot [34]; KungfuBot2 [35]; Expressive WBC [36]; SONIC [37]; Humanoid-GPT [38]; AMP [39]; NEAR [40]; SMP [41]; CMP [42]; ADP [43]; ASE [44]; C-ASE [45]; Universal Humanoid Motion Representations [46]; Hybrid Motion Priors [47]; Latent-Conditioned Loco-Manipulation [48]; ExBody2 [49]; HoRD [50]; VIRAL [51]; From Experts to a Generalist [52]; EAGLE [53]; X-LoCo [54]; BeyondMimic [55]; DAGger [56].
L3 Context-rich Interaction		VideoMimic [57]; HDMI [58]; OKAMI [59]; EgoHumanoid [60]; Humanoid Manipulation Interface [61]; Mobile-TeleVision [62]; DemoHLM [63]; OASIS [64]; SIMPLE [65]; HumanoidMimicGen [66]; InterMimic [67]; InterReal [68]; ContactMimic [69]; SceneBot [70]; WoCoCo [71]; ResMimic [72]; InterPrior [73].
L4 Multimodal task grounding		RT-2 [74]; π_0 [75]; OpenVLA [76]; WholeBodyVLA [77]; GR00T N1 [78]; Ψ_0 [79]; OpenHLM [80]; Being-0 [81]; MetaWorld-X [82]; Humanoid Policy $\sim$ Human Policy [83]; Behavior Foundation Model [84]; Scaling Behavior Foundation Model [85]; Meta Motivo [86]; TextOp [87]; LeVERB [88]; Commanding Humanoid by Free-form Language [89]; TANGO [90]; SENTINEL [91]; LangWBC [92]; ULTRA [93].
L5 Predictive decision making		DreamDojo [94]; UniT [95]; DreamZero [96]; World Action Planner [97]; ω -0 [98]; MotionWAM [99]; WholeBodyWAM (WBC-Grounded) [100]; WholeBodyWAM (Motion Priors) [101].

[19, 20, 102, 103]. A complementary route collects human motion through wearable trackers, motion-capture systems, VR/MR interfaces, or whole-body teleoperation, providing motion that is more closely aligned with the target robot and its control stack [21–23, 25]. In either case, raw human motion cannot usually be cloned to a humanoid directly because of differences in morphology, joint topology, degrees of freedom, and contact geometry. Retargeting therefore maps human motion into robot-compatible references, typically through inverse kinematics, contact-aware optimization, or learned morphology-aware mappings [27, 28]. However, even these retargeted trajectories specify only desired kinematics rather than deployment-ready references, and require additional physical grounding for robust real-world execution.

b) Reference-Based Motion Tracking: The most direct technique to ground the motion data in real world is to treat the retargeted trajectory as an explicit reference for reinforcement learning policies. An RL controller $\pi_\theta(a_t | o_t, r_t)$ is conditioned on a reference r_t , which may contain target joint poses, root states, end-effector states, or short-horizon motion information. The reference specifies *what* motion should be reproduced, while reinforcement learning learns *how* to execute it under the robot’s dynamics, balance requirements, contacts, and actuation limits. DeepMimic established this tracking-

based formulation for physics-grounded motion imitation [29]. Subsequent humanoid systems have expanded the same principle from individual skills toward broad families of natural and highly dynamic motions [30–35]. More recently, motion tracking has also become a scalable motor-pretraining objective. Systems such as SONIC and Humanoid-GPT increase both motion diversity and policy capacity to obtain controllers that generalize across large motion collections [37, 38]. In this sense, tracking has evolved from reproducing individual references into a mechanism for learning increasingly general whole-body motor competence.

c) Distributional Priors and Latent Motor Spaces: Motion data need not remain tied to a time-indexed reference. A more flexible approach summarizes demonstrations as a distributional prior over plausible motion. Rather than requiring the policy to follow a particular trajectory, adversarial, energy-based, or score-based objectives encourage its behavior to remain within the demonstrated motion distribution [39–43]. This distinction changes the role of the data: reference tracking specifies which trajectory to follow, whereas a motion prior specifies which behaviors are plausible or desirable while leaving the task objective greater freedom to select among them. Motion information can be compressed further into a reusable latent motor space. A low-level controller then maps a compact latent

TABLE II

SUMMARY OF LEARNING PARADIGMS FOR HUMANOID WHOLE-BODY CONTROL ALONG INCREASING DATA DIVERSITY AND TASK GENERALIZATION. THE LISTED YEARS INDICATE THE APPROXIMATE EMERGENCE OF EACH PARADIGM IN HUMANOID CONTROL RATHER THAN THE FIRST APPEARANCE OF THE UNDERLYING TECHNIQUES.

Category	Main methods	Core ideas	Year
L1 Reward-defined reactive control	On-policy and (emerging) off-policy RL [9–11]; massively parallel simulation, curriculum and privileged learning, and sim-to-real adaptation [1, 3, 4, 12]	Learn task-specific feedback policies from self-generated interaction. Low-dimensional commands and hand-designed rewards specify success, while simulation provides scalable experience for robust closed-loop control.	2016–
L2 Motion-conditioned control	Reference collection [18, 21–23, 25]; motion retargeting [27, 28]; RL grounding [29, 39, 44]; policy distillation [49, 52, 53, 55]	Use external human or robot motion data to specify desired behavior. Retargeting bridges embodiment gaps, while RL grounds reference trajectories, motion distributions, or latent motor priors in dynamically feasible closed-loop control.	2018–
L3 Context-grounded physical interaction	Demonstration collection [57, 62, 66]; context-rich alignment [58, 59, 61]; interaction-aware physical grounding [67, 71, 72]	Extend supervision from body motion to body–world relations by preserving object, scene, and contact context. The policy must realize both plausible whole-body motion and the intended physical interaction.	2024–
L4 Multimodal generalist control	Multimodal inference [74–76, 78]; robust real-world execution [37, 38, 84, 85]; skill, motion-latent, or direct-action coupling [77, 81, 88, 89, 91, 92]	Infer open-ended task intent from vision and language input, and then couple robust executor to enable physics-grounded whole-body execution. Multimodal models specify what to do, while behavior foundation models preserve robust high-rate control.	2025–
L5 Predictive world–action models	Predictive experience modeling [94, 95]; action-conditioned foresight and planning [96, 97]; prediction–execution coupling [98, 99]	Model how candidate actions change future observations, states, or latent world representations, adding explicit action-conditioned foresight. Prediction can guide a separate robust executor or be learned jointly with action generation.	2026–

variable z_t to executable whole-body actions, while an upstream policy selects continuous embeddings, discrete skill codes, or other motor representations instead of directly commanding all joints [44–48]. In summary, from explicit reference tracking to implicit motion distributions and latent motor manifolds, these formulations become progressively less tightly coupled to any particular demonstrated trajectory. This relaxation creates a natural trade-off: explicit references provide precise and interpretable motion commands but generalize poorly beyond the demonstrated trajectories, whereas implicit priors and latent representations offer greater behavioral flexibility and reuse at the cost of weaker trajectory-level specificity and less direct control over the realized motion.

d) Imitation Learning for Policy Distillation: Without the physical grounding mechanisms discussed above, policies learned through pure imitation learning remains difficult to deploy robustly on humanoids. Firstly, human motion data typically specify relatively high-level target kinematics rather than executable robot actions. Moreover, in high-dimensional, contact-rich, and balance-critical settings, even robot-native demonstrations provide limited closed-loop coverage, making supervised policies vulnerable to compounding distribution shift [56]. These limitations shift the role of imitation in humanoid control from directly acquiring robust behavior to transferring behavior that has already been physically grounded. In practice, this often takes the form of policy distillation. A teacher can first acquire dynamically feasible whole-body behavior through RL with privileged states, future references, or richer observations, and then transfer this competence through imitation learning to a student using only deployable sensing [23, 49–51]. Distillation can also merge multiple grounded specialists into a unified generalist policy [52–54], or transfer their behavior into more expressive policy representations, including large sequence models, diffusion policies, and multimodal controllers [38, 55, 93]. This staged transfer provides a practical scaling strategy: rather than optimizing a single policy over increasingly

broad and heterogeneous behaviors from scratch, specialized competence can be learned separately and then consolidated, enabling more stable data and capability scaling.

C. Context-rich Interaction

a) From Body Motion to Physical Interaction: Motion-conditioned control primarily describes how the humanoid body should move. A natural extension is from body motion alone to interaction data that additionally preserve object states, scene context, and contact relations. This distinction becomes essential in tasks related to human–object interaction and whole-body loco-manipulation. A motion may appear natural in isolation yet still fail if the hand approaches the wrong object pose, the support configuration is incompatible with the scene, or contact occurs at the wrong time. Context-rich control therefore specify not only *how the body moves*, but also *how that motion should relate to the surrounding world* [57–61].

b) Demonstration Collection and Alignment: In terms of demonstration data, L3 largely inherits the motion-acquisition interfaces of L2, but augments body motion with the physical context required for interaction. Such context can be collected from exocentric human videos that jointly recover body, object, and scene interaction [57–59]. It can also be obtained from robot-free human operator interfaces that capture task-space trajectories together with egocentric observations [60, 61]. Finally, some whole-body teleoperation systems also jointly provide robot-native proprioception, vision, and control signals which are useful for contact-rich task training [23, 62]. What’s more, these data can then be augmented in simulation by adapting source demonstrations to new object configurations and scene layouts through trajectory transformation and whole-body planning [63–66]. In summary, the resulting supervision extends plain body-motion trajectories into synchronized descriptions of the humanoid, surrounding objects, scene context, and their physical interaction.

Converting such demonstrations into robot supervision requires alignment at several levels in addition to conventional motion retargeting. First, *embodiment alignment* maps reconstructed human motion to the target humanoid through task-space retargeting and inverse kinematics [58, 59]. Second, *interaction-context alignment* preserves object-relative motion and contact structure through object-centric coordinate frames, trajectory warping, and explicit contact supervision [58, 59, 63], allowing demonstrations to remain meaningful across different object poses and scene configurations. Finally, *cross-embodiment alignment* brings human and robot data into compatible observation and action spaces, either through shared task-space representations and viewpoint transformation or through human-in-the-loop adaptation during collection [60, 61]. Context alignment thus extends the L2 mapping from human to robot morphology into a joint alignment of body motion and physical interaction.

c) Interaction-Aware Physical Grounding: Many interaction-control methods extend the physical-grounding machinery introduced in L2, including reference tracking, motion-prior regularization, and latent motor policies trained with RL. L3 augments this machinery with object or scene states, body–environment relations, and contact information in the policy inputs, references, or learning objectives. Representative strategies follow three recurring patterns. First, interaction-aware conditioning and objectives incorporate object tracking, relative geometry, and explicit contact signals so that the policy optimizes not only body motion but also its relation to the environment [67–70]. Second, long-horizon contact-rich tasks can be structured around sequential contact stages to reduce exploration difficulty and simplify task specification [71]. Third, a motor policy pretrained on broad human motion can be retained as a reusable backbone, while an object-conditioned residual specializes it for precise interaction and contact execution [72]. Together, these strategies extend the L2 objective of learning *how the body should move* toward learning *how that motion must adapt to body–world interaction requirements*.

D. Multimodal task grounding

a) From Structured Tasks to Multimodal Intent: Context-grounded interaction in L3 substantially broadens the range of physical behaviors a humanoid can acquire, but the task itself is still typically specified through a predefined object, goal, reference interaction, or reward. Multimodal generalist control relaxes this assumption by allowing task intent to be inferred from visual observations and open-ended language. This direction largely inherits from vision–language–action (VLA) models in manipulation, where large-scale vision–language pretraining and heterogeneous robot demonstrations enable policies to connect semantic concepts with executable actions [74–76]. Recent humanoid systems extend the same idea toward whole-body loco-manipulation and general-purpose control [77–80]. Along κ_d , the important change is therefore not simply an increase in demonstration volume, but a shift from context-rich physical interaction data toward heterogeneous multimodal experience in which the desired task can itself be specified according to multimodal understanding.

b) Inference–Execution Decomposition: Early manipulation-side VLAs tightly coupled multimodal understanding and action prediction within a unified sequence model, for example by representing robot actions as tokens predicted alongside language [74, 76]. Based on this, more recent architectures increasingly specialize these functions internally, pairing a pretrained vision–language backbone with a dedicated continuous-action expert that can be jointly trained or introduced during task-specific post-training [75]. Both designs nevertheless remain end-to-end in action mapping, with no explicit hand-designed skill, subgoal, or motion interface exposed between multimodal input and robot action.

For humanoid whole-body control, such architectural separation becomes increasingly necessary because multimodal inference must coexist with more demanding action generation involving high-dimensional actuation, floating-base balance, discontinuous contacts, and substantially faster feedback requirements. Accommodating these requirements within a unified architecture is considerably more difficult. A common design therefore further separates multimodal inference from physics-grounded execution into modular components: the former interprets scene context and task intent, while the latter realizes that intent through a high-rate closed-loop motor policy [77, 81, 82]. In many systems, this decomposition extends to the training pipeline, allowing multimodal understanding and whole-body physical grounding to be learned from different data sources and at different stages.

This decomposition reveals two complementary directions for scaling humanoid foundation models. *Multimodal scaling* broadens the range of embodied intent that can be inferred through stronger multimodal backbones, language supervision, human egocentric video, heterogeneous robot experience, and synthetic data [78–80, 83]; *Robust motor scaling* broadens the range of behaviors that can be executed robustly by increasing motion diversity, interaction experience, and policy capacity. Recent works have shown that large-scale motion tracking policies introduced in L2 can be naturally turned into reusable whole-body motor priors [37, 38, 84–86]. In summary, current progress can therefore be viewed as jointly scaling *what the humanoid can understand* and *how it can physically execute*.

c) Coupling Multimodal Intent to Whole-Body Execution: The remaining question is how multimodal intent is coupled back to the motor layer. At the modular end, a high-level model decomposes instructions into subgoals or selects reusable motor skills that are executed by separately trained controllers [81, 82]. This design naturally isolates high-frequency whole body action coordination from low-frequency multimodal inference, and preserves simplicity and reusability. However, the predefined skill library can restrict task generalization, and its low-dimensional high-level interface may be poorly aligned with downstream whole-body feasibility.

Besides explicit interfaces, closer coupling replaces discrete skills with generated reference trajectories, motion tokens, or continuous latent commands [77, 87–90]. Such interfaces expose a richer behavior space to multimodal reasoning while retaining a lightweight, execution-aligned interface to a physics-grounded executor, providing a practical compromise between generalizable multimodal understanding and robust high-rate

control.

At the tightest end, several works have proved that humanoid systems can still retain the pattern of coupling a pretrained multimodal backbone with a dedicated action expert while removing any externally exposed skill or motion interface, which is favored by the current manipulation VLA systems [78–80, 91, 92]. These models may still contain internal specialization or a learned low-level stabilizer, but the semantic-to-action pathway is trained as a tightly coupled system through robot-action post-training, heterogeneous co-training, RL-based grounding, or distillation. This tighter coupling reduces interface-induced information loss and can enable more task-dependent whole-body coordination, but places substantially greater demands on action data, optimization, real-time inference, and safety.

E. Predictive Decision Making

a) From multimodal Understanding to Predictive Foresight: Multimodal foundation models substantially expand the ability of robots to interpret open-ended tasks, but they remain primarily grounded in the present observation and the desired intent. A natural next step is to ask whether a robot can anticipate the physical consequences of its actions before execution. Predictive world-action models (WAMs) address this question by learning action-conditioned regularities of future observations, states, or latent world representations. Rather than only mapping multimodal context to actions, they aim to model action-conditioned future evolution by predicting how observations, states, or latent world representations change after an intervention. In this sense, WAMs extend multimodal understanding with physical foresight: the model not only reasons about what should be done, but also predicts what may happen after acting. However, predictive world modeling remains an emerging direction in humanoid control, rather than the dominant paradigm, due to the difficulty of collecting large-scale action-conditioned embodied experience.

b) Learning Predictive World–Action Models: Unlike conventional robot learning datasets that mainly contain individual observations and actions, WAM training requires temporally structured experience describing both interventions and their consequences. One source of such experience is large-scale passive video, including human and robot interaction videos, which provides broad physical priors despite lacking explicit action labels. DreamDojo converts such video experience into useful representations through latent action discovery and video–action alignment [94]. A second source is action-conditioned robot experience, where synchronized observations, proprioception, and actions directly expose controllable world dynamics. DreamZero jointly models future video and actions for closed-loop interaction, while World Action Planner explicitly exploits action-conditioned predictions for decision making [96, 97]. For humanoids, ω -0 learns predictive representations from embodiment-specific multimodal experience [98], while WholeBodyWAM introduces a complementary route by pretraining future whole-body motion from large-scale heterogeneous human and humanoid data before target-robot action learning [101].

c) Coupling Prediction with Physical Execution: Predictive models do not eliminate the need for a physically grounded controller; instead, they provide additional foresight for decision making and motion generation. One emerging paradigm is to use prediction as an intermediate planning representation, where future world predictions are converted into motion objectives or latent commands that are executed by a robust motor policy. For example, MotionWAM couples predictive visual representations with motion generation for downstream whole-body execution [99], while WholeBodyWAM connects pretrained world–action priors to heterogeneous controllers through a structured WBC interface [100], both preserving low-level physical grounding. Another direction tightly integrates prediction and action generation within a unified model, as explored by DreamZero and ω -0 [96, 98]. Related work such as WholeBodyWAM instead strengthens this trend through large-scale pretraining of predictive world–action priors for humanoid control [101]. Even in these more integrated architectures, embodiment-specific stabilization, contact handling, and feedback control remain essential. Thus, predictive world-action models should be viewed as an additional layer of foresight above existing motor foundations rather than a replacement for physics-grounded execution.

While current humanoid systems still rely primarily on reward learning, motion priors, and multimodal task grounding, predictive world-action models represent a further step toward agents that not only understand tasks and execute behaviors, but also anticipate the physical consequences of their decisions.

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